

Lecture 随机策略策略.

⇒ 为何确定性策略不可能成功? question

该确定性策略, 其问题在于 ⇒ 对手会知道你的策略.

即对手可以反向设计, 且任何固定规则均存在漏洞.

所以任何确定性策略在面对一个足够且对抗性的对手时, 都会被击败! KE!

"mcs" — 2015/5/18 — 1:43 — page 748 — #756

⇒ 关于随机策略, 其利用了对方无从预判的性质.

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Chapter 18 Random Variables

1. 即使随机切分点太密, 那只是退回均匀分布, 没有损失.

2. 且随机性能保证“不可能被完美针对”.

⇒ 用博弈论来总结: 你的策略只是一个概率分布, 因此对手必须应对所有可能性, 从而平均下来得到一个正的期望收益.

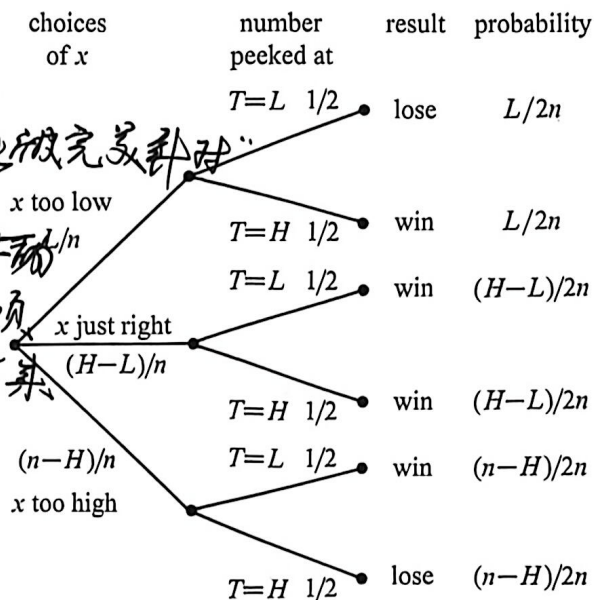


Figure 18.3 The tree diagram for the numbers game.

Step 4: Compute event probabilities.

The probability of the event that you win is the sum of the probabilities of the four outcomes in that event:

$$\begin{aligned} \Pr[\text{win}] &= \frac{L}{2n} + \frac{H-L}{2n} + \frac{H-L}{2n} + \frac{n-H}{2n} \\ &= \frac{1}{2} + \frac{H-L}{2n} \\ &\geq \frac{1}{2} + \frac{1}{2n} \end{aligned}$$

The final inequality relies on the fact that the higher number H is at least 1 greater than the lower number L since they are required to be distinct.

Sure enough, you win with this strategy more than half the time, regardless of the numbers in the envelopes! So with numbers chosen from the range $0, 1, \dots, 100$, you win with probability at least $1/2 + 1/200 = 50.5\%$. If instead we agree to stick to numbers $0, \dots, 10$, then your probability of winning rises to 55% . By Las Vegas standards, those are great odds.